***LitE*: Load Balanced Virtual Data Center Embedding for Energy Efficiency in Data Centers**

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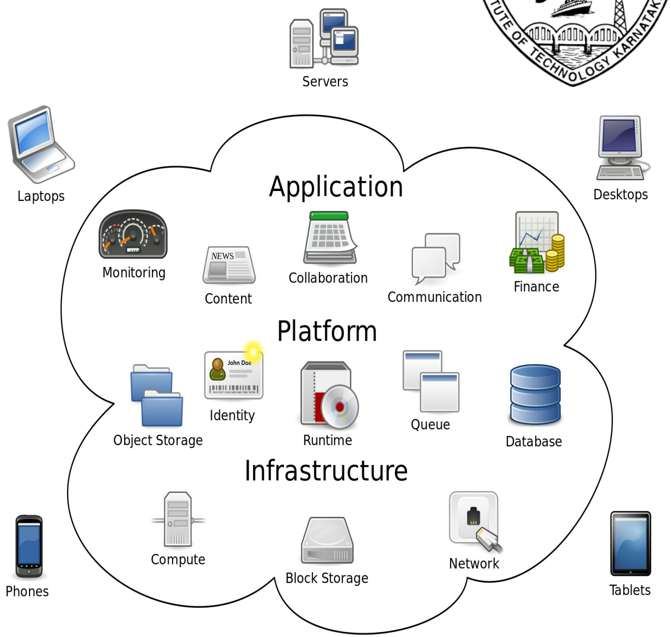
CLOUD AND SMART SYSTEM SERVICES

Cloud Computing

- Cloud computing enables users to access shared resources, software, and information over the internet from any device.

- This model offers scalable and flexible IT services, allowing businesses to utilize powerful computing capabilities without extensive on-premises infrastructure.

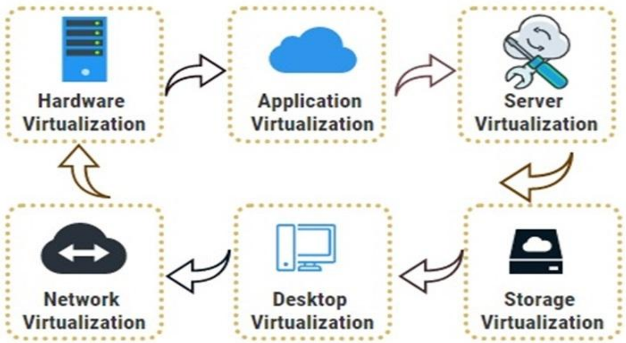


Source: A. AWS, “Cloud Computing,” https://s3.amazonaws.com/wp-content/uploads/2016/03/Cloud computing.png, 2024.

Virtualization

The technology allows us to create multiple virtual environments or resources from a single physical system.

- It separates the hardware from the software, enabling different systems or applications to share the same resources efficiently.
- This technology is widely used to improve efficiency and flexibility in managing computing.



Source: Educba, "types of virtualization," https://cdn.educba.com/academy/wp-content/uploads/2019/06/Types-Of-Virtualization.png, 2024.

Network virtualization (NV)

NV is a technology that abstracts and combines network resources and functionality into a single, software-based administrative entity known as a Virtual Network (VN).

- NV is set to enable effective resource management in next-generation Internet architectures by logically separating network services from the underlying hardware.
- It allows for creating multiple isolated VNs that can share the same underlying physical network (PN) infrastructure.
- This abstraction simplifies network management, increases flexibility, and enhances the efficiency and scalability of network resources.

Virtual Network (VN)

VN are software-based networks that emulate PNs by abstracting resources like switches, routers, and servers, enabling flexible and scalable management without altering the physical infrastructure.

Key components of VN

- Virtual Nodes are software-based components replicating physical network devices, enabling data routing and management within a virtual network.
- Virtual Links are software-based connections that replicate physical network cables, enabling data transmission between virtual nodes within a virtual network.

Virtual Data Center Embedding (VDCE)

In NV, VDCE is the process of allocating Virtual Data Center Requests (VDCRs) to physical resources, a process which is NP-hard.

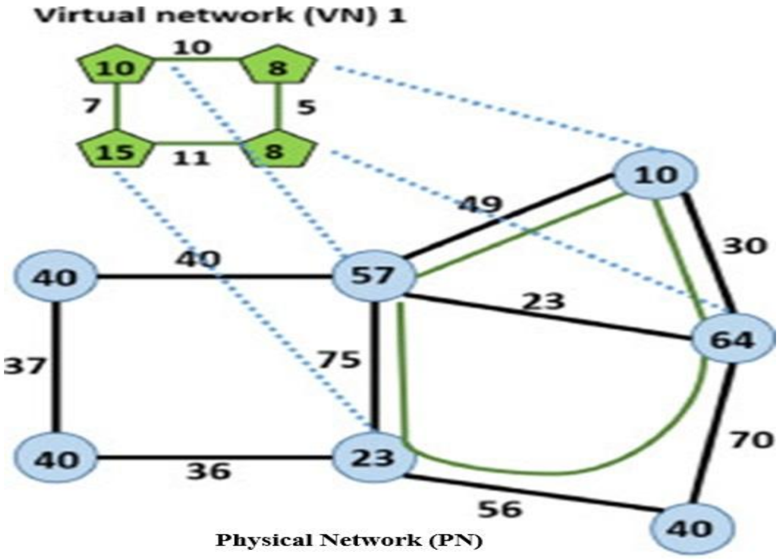
➔ VDCRs are tasks or applications requested by end users.

Key Concepts of VDCE

- VDCRs consist of virtual nodes (e.g., virtual machines or virtual routers) and virtual links (e.g., virtual connections between nodes). These virtual components need to be mapped onto the physical infrastructure.
- Physical Network (PN) comprises physical nodes (e.g., physical servers, switches) and physical links (e.g., physical cables, wireless links). The PN provides the underlying resources for hosting VNs.

Virtual Data Center Embedding (VDCE) *(contd.)*

VDCE involves two sub-problems: Virtual Machine (VM) embedding, assigning physical resources to VMs, and Virtual Link (VL) embedding, mapping physical paths to VLs connecting the VMs.

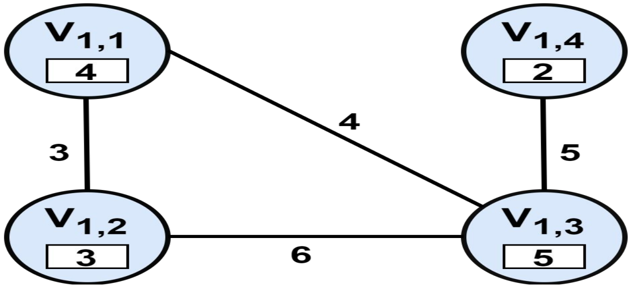


Mapping VN into traditional PN

Virtual Data Center Embedding (VDCE) *(contd.)*

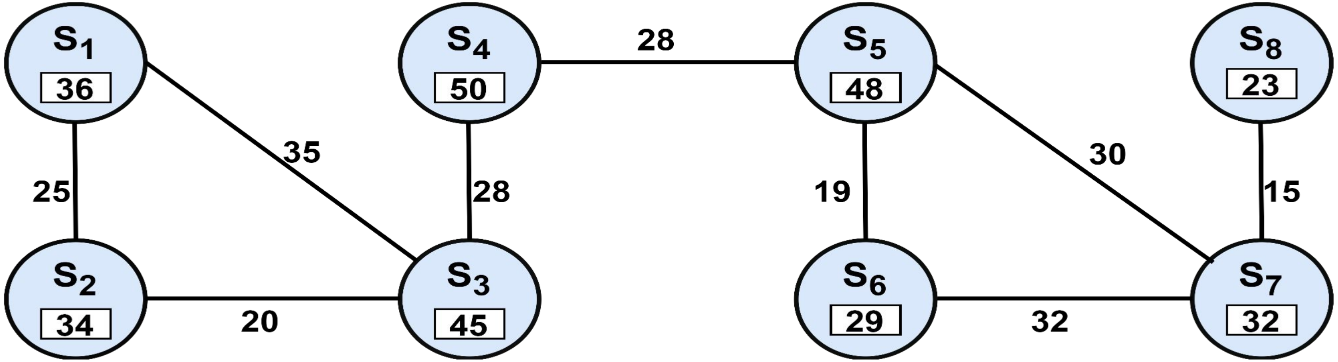
- Below Fig. represents the VDCR belonging to a sample of real-time applications such as hosting websites, online games, and video streaming from geographically distributed users.
- Below VDCR comprises four VMs, and four interconnected VLs. The numerical value 4 associated with VM “$V_{1,1}$” represents its resource demand in terms of computational resource blocks (CRBs).
- One CRB unit equals one CPU core and 512 MB of RAM. Similarly, the numerals associated with VLs represent the communication bandwidth demand.

An instance of a virtual data center request



Virtual Data Center Embedding (VDCE) *(contd.)*

- ➔ The sample traditional PN shown in below Fig. comprises eight physical servers and nine interconnected physical links (PLs).
- ➔ The numerical value 36 associated with server "S_1" represents its resource available in terms of CRBs. Similarly, the numerals associated with PLs represent the communication bandwidth available.



An instance of a traditional physical network

Challenges in VDCE

- Choosing the best method for VDCR embedding: Deciding between centralized or distributed methods for embedding VDCRs and managing network resource data.
- Identifying the appropriate solution approach for the VDCE process: Determining the most effective and efficient approaches for embedding VDCRs into the PN.
- Challenge in incorporating real-time adaptability: Effectively integrating real-time adaptability in VDCE remains difficult, leading to suboptimal embedding decisions and inefficient resource utilization.
- Conducting tests using commercial simulator providers presents challenges due to the scale of the problem and associated costs.

Motivation

- While the research community has focused on formulating embedding strategies, there has been less emphasis on practical implementation at a laboratory scale, which is crucial for comprehensive design, development, testing, and validation policies for large-scale systems.
- Most VDCE solutions focus on system resources, neglecting real-time network topology, which leads to less effective outcomes.
- Moreover, current open-source simulators lack accuracy in representing the complexities of communication patterns, resource allocation.
- These limitations result in inefficient implementations and reduced adaptability, hindering seamless integration with commercial cloud providers.

Concerns by Cloud Service Providers (CSPs)

A major concern by CSPs is the high energy consumption in DCs, compounded by low resource utilization rates.

- On average, DCs operate at only 15-20% CPU usage, resulting in substantial waste, as idle physical machines (PMs) consume up to 70% of their peak energy.
- Globally, the Information and Communication Technology (ICT) infrastructure consumes approximately 10% of the world's energy, with U.S. DCs alone consuming 1.4% of national electricity in 2010 and 1.8% in 2014.
- Projections suggest that IT energy consumption could reach 13% by 2030, with DC electricity use growing by 15-20% annually.
- Further, Amazon reports that 42% of its operational costs are due to DC energy use.

Source: 1. M. H. Ferdaus, M. M. Murshed, R. N. Calheiros, and R. Buyya, "Multi-objective, Decentralized Dynamic Virtual Machine Consolidation using ACO Metaheuristic in Computing Clouds," CoRR, vol. abs/1706.06646, pp. –, 2017. [Online]. Available: http://arxiv.org/abs/1706.06646.
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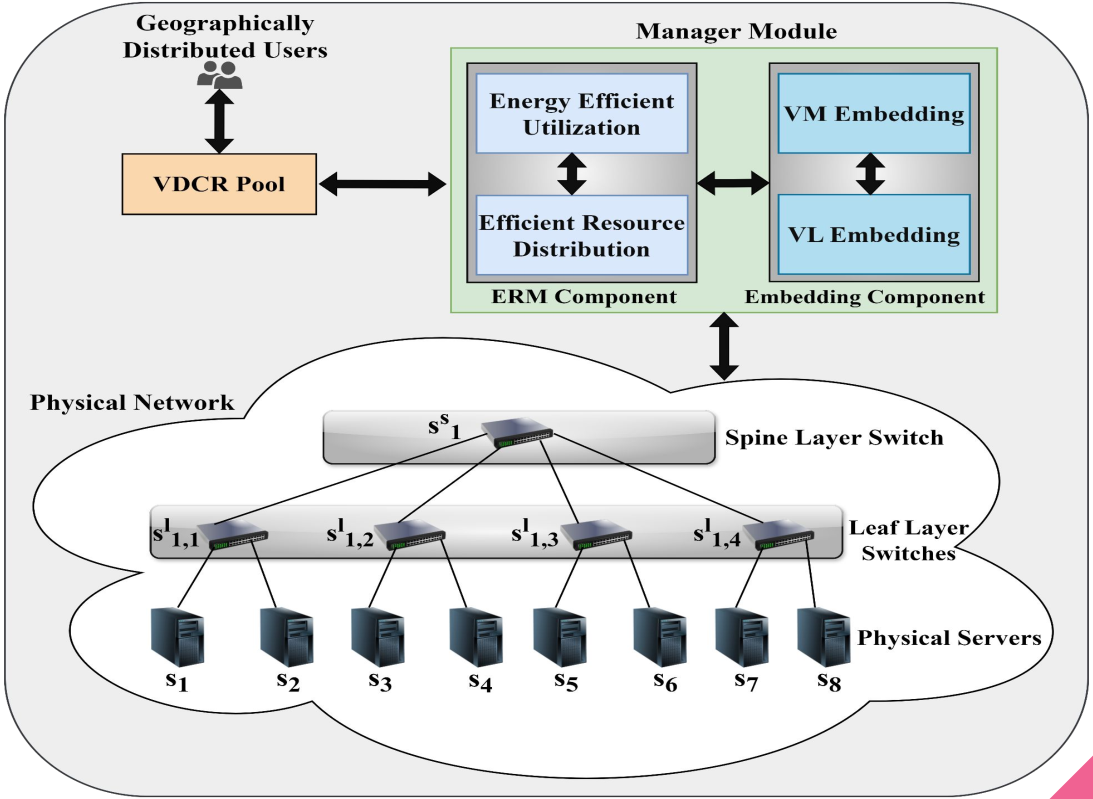
Contributions

The key contributions of this work are as follows:

- This work introduces a framework called LitE for the VDCE problem. It generates a spine-leaf topology-based PN, typically called a full-meshed Clos architecture.
- The proposed work aims to minimize the overall energy consumption in DC through effective load-balancing and thereby improve the overall performance of the network.
- LitE offers an Efficient Resource Management (ERM) component to evaluate the VM embedding benefits of the server by considering server utilization, server overloading probability, and server energy consumption.
- Using this embedding data, VM embedding is carried out, followed by a VL embedding using Dijkstra's shortest path algorithm.
- To test LitE effectiveness, we integrated three state-of-the-art VDCE strategies (i.) Congestion-Aware, Energy-Aware Virtual Network Embedding (CEVNE),(ii.) Dynamic Region of Interest (DROI) and (iii.) First Fit algorithm.

CLOUD AND SMART SYSTEM SERVICES

Proposed *LitE* System Model



System Model *(contd.)*

Virtual Data Center Request (VDCR)

- Pool of VDCRs : $\mathbf{G}_{\boldsymbol{v}}$= {$G_1$,$G_2$, . . . ,$G_i$, . . .}.
- Where $G_i \in G_v$ is represented as an undirected weighted graph.
- G_i=(N_i , L_i), where N_i={$v_{i,1}$, $v_{i,2}$, . . .} is the set of VMs.
- L_i={$e^i_{1,1}$, $e^i_{1,2}$, . . . , $e^i_{j,j'}$, . . .} is the set of VLs.

Physical Network (PN)

- Spine-leaf topology-based PN : $\mathbf{G}_{\boldsymbol{p}}$=($N_p$, L_p).
- N_p=$N_s \cup N_r$, where N_s= {s_1, s_2, s_k, . . .}, and N_r ={N_{sl} ,N_{ll} }.
- L_p= {e_1, e_2, e_l , . . .} set of physical links.

Manager Module

- This module comprises the ERM and embedding components.
- The ERM is responsible for determining the most suitable server for each VM, focusing on two key factors: (i.) Energy-efficient utilization and (ii.) Efficient resource distribution.
- We start with estimating the energy consumption of each server. This estimation is based on the server's idle and full energy levels, scaled by its resource utilization (U_k).

$$\mathbb{E}_k(\mathbb{U}_k) = \mathbb{E}_k^{\text{idle}} + \left(\mathbb{E}_k^{\text{full}} - \mathbb{E}_k^{\text{idle}}\right) \cdot \mathbb{U}_k \quad \text{where,} \quad \mathbb{U}_k = \frac{\sum_i^{|\mathbb{G}_v|} \sum_j^{|\mathbb{N}_i|} r_{i,j}}{c_k^a}$$

- The second aspect is assessed through the server overloading probability and is computed using the resource's mean and standard deviation across server's in the DC.

$$\mathbb{P}(s_k) = 1 - \Phi\left(\frac{c_k^{\mathrm{r}} - r_{i,j} - \mu}{\delta}\right)$$

- The ERM component considers a server with minimum energy consumption and minimum overloading probability to ensure that the most suitable server is chosen for VM placement. Subsequently, the embedding component leverages server information from the ERM to perform two-stage VM embedding followed by a VL embedding using Dijkstra's shortest path algorithm.

$$\Gamma_{\mathrm{k}} = \mathbb{E}_k(\mathbb{U}_k) \cdot e^{\alpha \cdot \mathbb{P}(s_k)}$$

Objectives of *LitE*

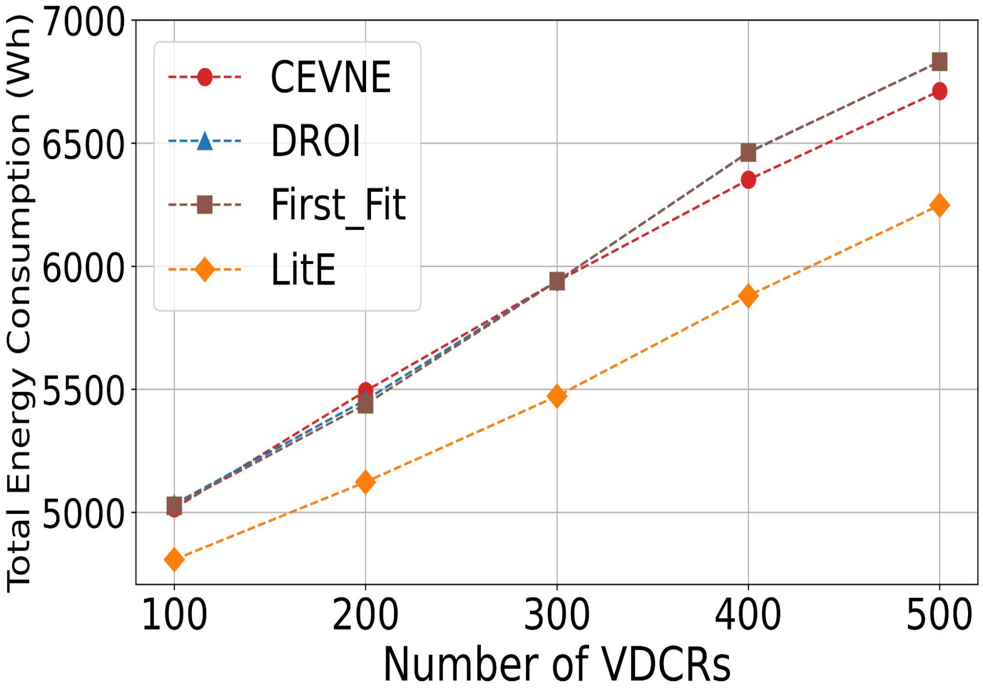
Problem Formulation

- **Objectives**
  - Minimization of Energy Consumption
  - Load Balancing
- **Constraints**
  - VMs and VLs CRB demands do not exceed available resource limits
  - Node indicator's and Path indicator's are set to 1 if VM $v_{i,j}$ is assigned to server s_k and Y($v_{i,j}, s_k$) & Y($v_{i,j'}, s_{k'}$) is assigned to physical link, otherwise 0
  - No VMs from the same VDCR are placed on the same server

Testbed Setup

- We have performed simulations using Mininet.
- A single-domain spine-leaf topology-based PN with 30 servers and 36 links is used in LitE.
- The resource capacities of the servers are uniformly distributed in the range U[200, 1000].
- The bandwidth link capacities from server to leaf switches and from leaf switches to spine switches (in Gbps) are within the range U[500, 1000].
- Each VDCR has VMs ranging from U[2, 10], with a link connectivity probability of 0.4, and VDCRs arrive at Poisson-distributed λ=0.4.
- The VM resource demand and VLs bandwidth demand are uniformly distributed within the range U[1, 10] and U[1, 5], respectively.
- We have considered five scenarios of VDCRs [100, 200, 300, 400, 500], each running ten times.
- Experiments ensured a 100% acceptance ratio as in to demonstrate the energy consumption across the test scenarios.

RESULTS Energy Consumption

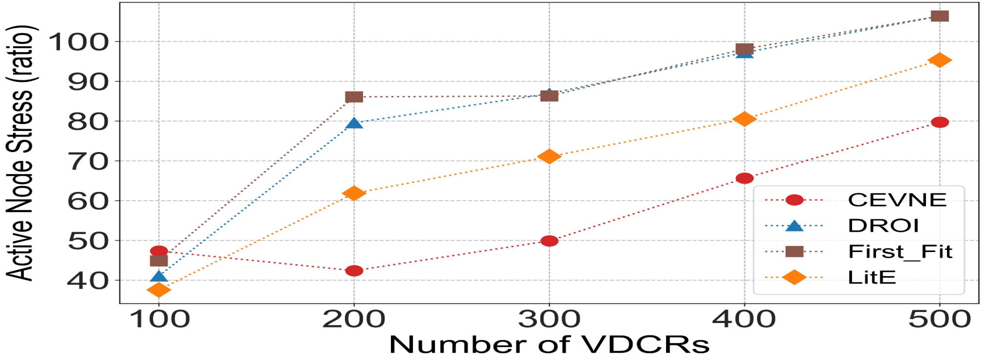


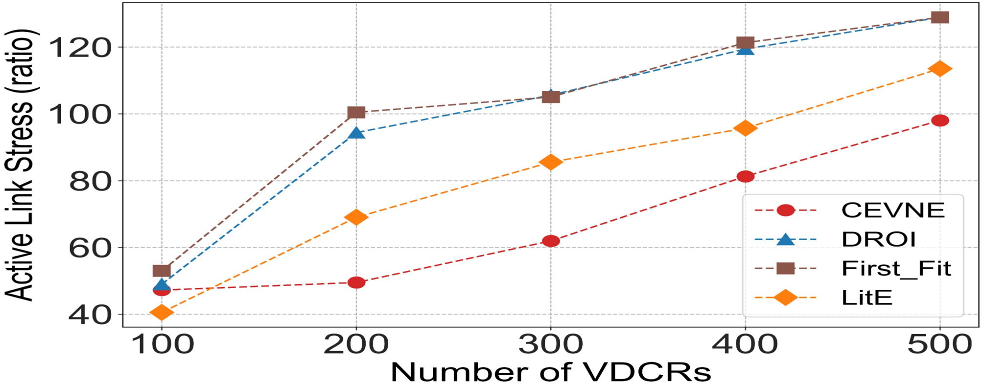
LitE consistently utilizes the least energy and outperforms the baselines such as First Fit, DROI, and CEVNE. This is due to LitE server selection mechanism based on resource utilization and energy metrics, which prevent high-energy spikes through effective load balancing.

LitE minimizes the energy consumption by 15% compared to baselines.

RESULTS

Active Node and Link Stress

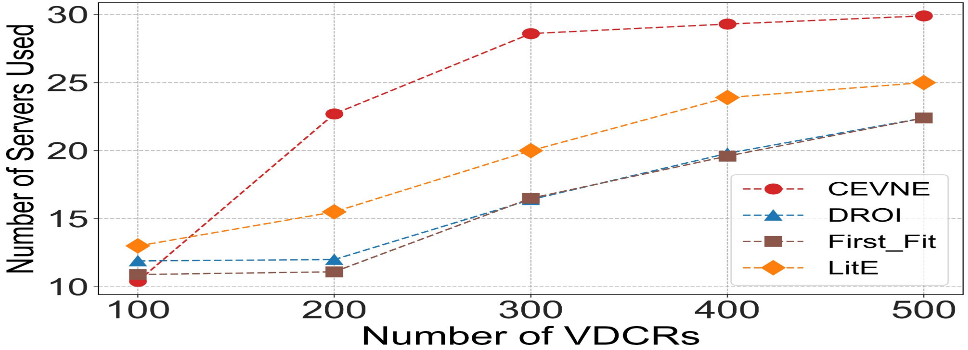


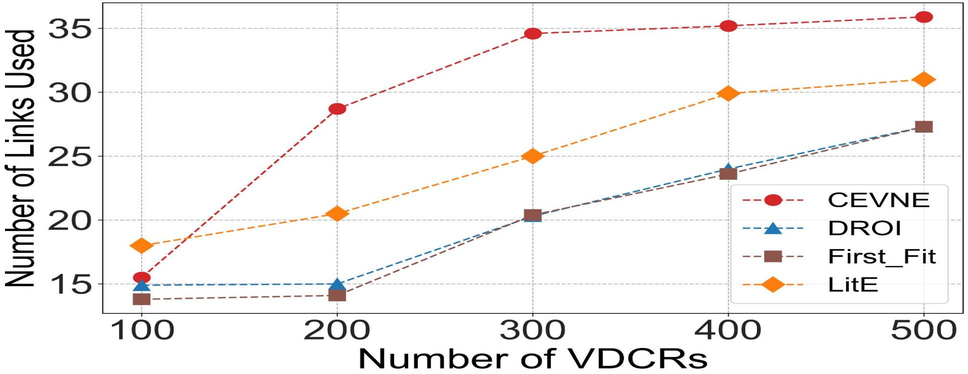


LitE achieves moderate node and link stress by considering server utilization during VM embedding and balancing network load, which distributes resources more evenly across servers and links. In contrast, DROI and First Fit cause higher node and link stress due to uneven distribution and poor server selection, leading to imbalance. CEVNE maintains the lowest node and link stress but risks underutilizing server resources and physical links, resulting in lower overall performance.

RESULTS

Servers and Links Used

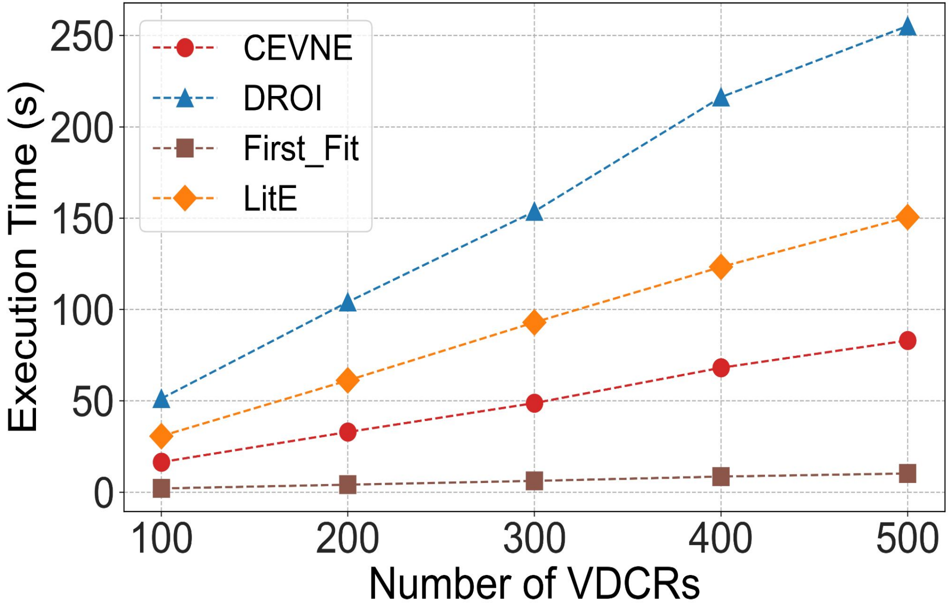




LitE uses fewer servers than CEVNE but slightly more than DROI and First Fit by optimizing server allocation based on resource overloading probability for efficient resource usage. It also evenly distributes the network load, maintaining stable physical link usage and improving performance. In contrast, CEVNE uses more links, causing inefficiencies, while DROI and First Fit use fewer links, increasing node stress, energy consumption, and potentially lowering performance.

RESULTS

Execution Time



LitE takes moderate time than CEVNE and First Fit because it focuses on finding load-balanced and energy-efficient servers during VM embedding. In contrast, First Fit has the fastest execution time due to its simple server selection strategy, while DROI experiences a sharp increase in execution time due to its complex clustering mechanism.

Conclusion

- ➔ This work presents the LitE framework, a VDCE approach that minimizes energy consumption by balancing load in a physical network.
- ➔ LitE considers server resource utilization and overloading probability during VM embedding and uses the shortest path algorithm for VL embedding.
- ➔ This improves energy efficiency and resource management. Simulation results show that LitE reduces data center energy consumption by 15% compared to baseline methods.

Future Work

Integrate LitE with software-defined networking features to test its performance in dynamic environments and validate it on commercial cloud platforms like AWS and Azure.

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Thank You

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